

DATE	30 September 2026
TEAM ID	SWUID2026349965
PROJECT NAME	PocketSmart AI: Your Smart Budget & Recommendation Assistant

Phase 07: Project Documentation

1. Project Summary

PocketSmart AI is a GenAI-powered budget and recommendation assistant. Users enter a budget and preferences for home interiors, parties or jewelry and receive a Gemini 2.5 generated plan with estimated prices and shopping links for Indian platforms.

1.1 Key Features

- Three planners: Home Interior, Party and Jewelry (with optional outfit image).
- Cross-platform links: Amazon, Flipkart, IKEA, Swiggy, Zomato, OYO, BookMyShow, Bluestone, Tanishq and more.
- Secure login with hashed passwords, access tokens and session tracking.
- History of all plans with a full detail view.
- Responsive card-based interface.

1.2 Technology Summary

Area	Choice
Language and framework	Python, FastAPI, Uvicorn
AI	Google Gemini 2.5 through the Google Generative AI SDK
Frontend	HTML, CSS, JavaScript, Jinja2 templates
Security	bcrypt password hashing, JWT access tokens

2. Installation and Setup Guide

2.1 Prerequisites

- Python 3.10 or newer with pip.
- A Google AI Studio account and a Gemini API key.
- Visual Studio Code (recommended).

2.2 Setup Steps

Step	Action
1	Copy the project folder to your computer and open it in Visual Studio Code.
2	Create and activate a Python virtual environment.
3	Install the required packages: FastAPI, Uvicorn, python-multipart, Jinja2, Pillow, Pydantic, email-validator, passlib with bcrypt, python-jose, python-dotenv and the Google Generative AI SDK.
4	Create a file named .env in the project folder and add GOOGLE_API_KEY and SECRET_KEY.
5	Run the application file with Python (or start it with Uvicorn).
6	Open a browser and go to http://localhost:8000.

2.3 Environment Variables

Variable	Purpose
GOOGLE_API_KEY (or GEMINI_API_KEY)	Authenticates requests to Gemini 2.5
SECRET_KEY	Signs the access tokens

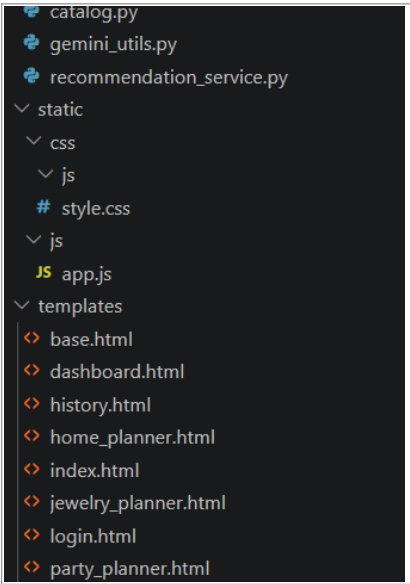


Figure 1: Project folder structure

3. User Guide

3.1 Getting Started

- Open the website, click Get Started and create an account.
- Log in to reach the dashboard and choose a planner.

Planner	How to use
Home	Enter the budget, number of lights, fans, furniture pieces and dining tables, tick the rooms, add notes and click Generate Recommendations.
Party	Enter the budget, guests, party type and venue type, select the needs, add notes and click Generate Budget Plan.
Jewelry	Enter the budget, occasion and style, optionally upload an outfit photo and click Get Recommendations.

3.2 Reading a Result

- Budget summary: total budget and remaining amount.
- Category cards: allocation with item name, description, price and quantity.
- Shopping links: click a platform button to search for that item.
- Suggestions and tips: ways to save money or style the items.

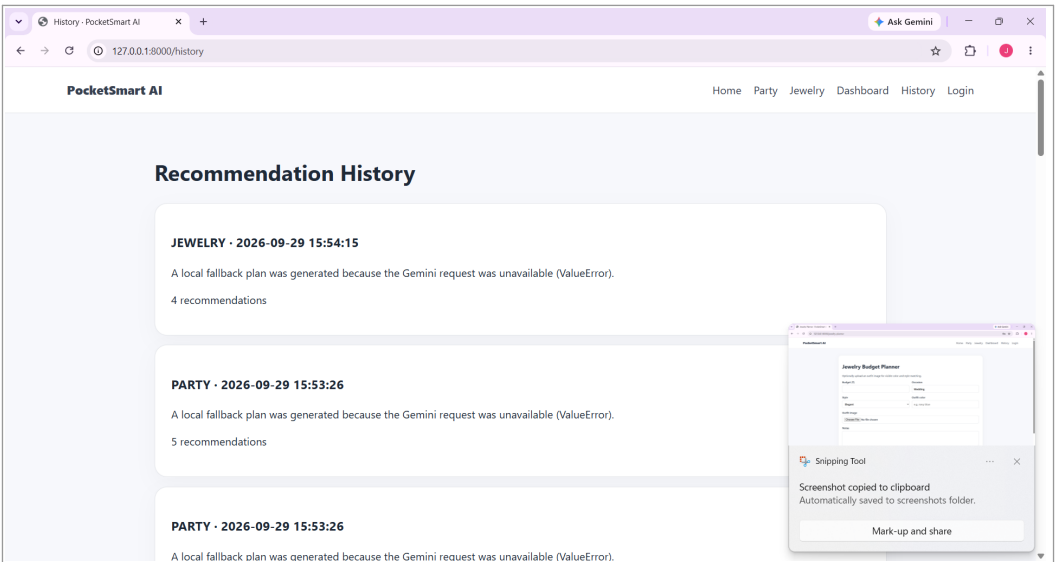


Figure 2: History page for reviewing earlier plans

4. Reference, Troubleshooting and Future Scope

4.1 Main Routes

Route	Method	Purpose
/token	POST	Log in and receive an access token
/home-budget	POST	Home plan
/party-budget	POST	Party plan
/jewelry-budget	POST	Jewelry plan with optional image
/recommendation-history	GET	List saved plans
/recommendation-details/{id}	GET	Full details of one plan
/session-info	GET	Session information

4.2 Troubleshooting

Problem	Solution
No Google API key error at start	Add GOOGLE_API_KEY to the .env file and restart
Error generating recommendations	Check the internet connection, API quota and model name, then try again
Login returns to the login page	The session expired after 30 minutes; log in again
Image not accepted	Use a JPG, PNG or WEBP image within the size limit
Port 8000 already in use	Start the server on another port

4.3 Known Limitations

- Prices are AI estimates, not live prices.
- Users, sessions and history are kept in memory in the demonstration build.
- Shopping links are searches, not exact product pages.

4.4 Future Scope

- Database storage and AWS deployment.
- Live price integration, fashion and travel planners, regional languages and voice input.
- Shared group budgets and export of plans.